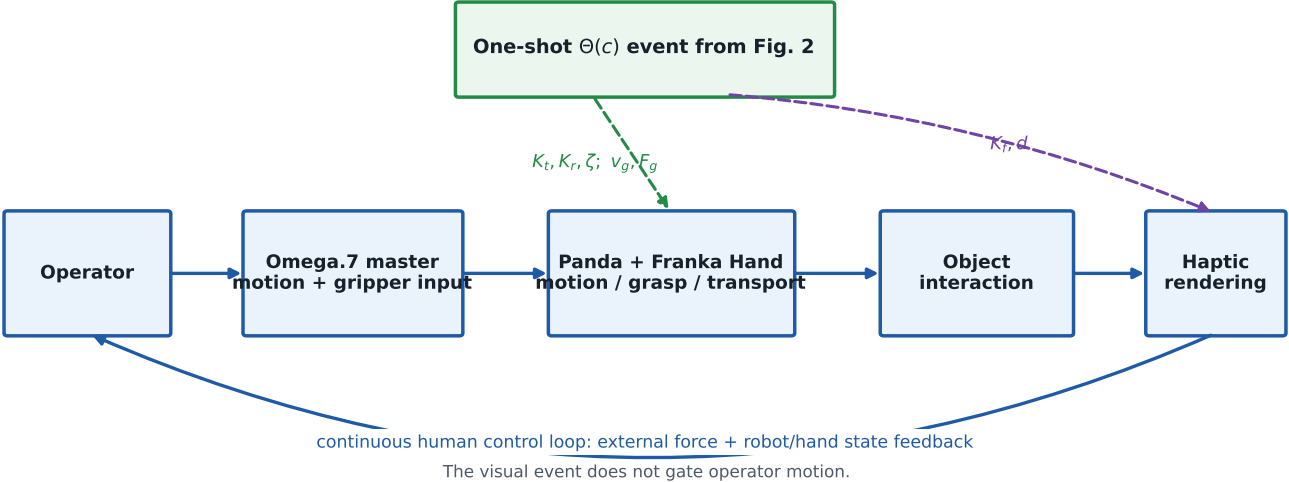


(a) Human-in-the-loop teleoperation and parameter injection



(b) Five-mode ablation matrix

Mode	Information / selection	Slave impedance K_t, K_r, ζ	Haptic interface K_r, d	Gripper execution v_g, F_g
A	Fixed parameters	—	—	—
B	Operator-selected	✓	✓	✓
C	Vision-semantic	✓	✓	✓
D	Visual cue only	—	—	—
E	Vision-semantic	✓	—	—

Checks denote strategy-dependent parameter updates; Table 3 gives initialization and fallback details.